

PRASHANT YADAV

Lucknow, Uttar Pradesh, India

ydvprashant0508@gmail.com | linkedin.com/in/pr7nt | github.com/YdvPrashant

SUMMARY

Software engineer working across full-stack web and applied ML. Deployed a Next.js news-transparency platform whose fact-check engine measures 93.3% with zero incorrect verdicts, plus an offline CLIP photo-search app in Python.

EDUCATION

Lovely Professional University

Phagwara, Punjab, India

Bachelor of Technology in Computer Science Engineering

July 2025

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems

PROJECTS

Prism: News Transparency & Bias Analysis Platform

Live

Next.js, TypeScript, React, Tailwind, Node.js, LLM APIs (Gemini, Groq), RAG

April 2026 – Present

- Built and **deployed** a news-transparency web app that refracts any article URL, YouTube link, or pasted text into a six-stage analysis: sentence-level claim/opinion/rhetoric classification, source provenance, live fact-checking, and coverage gaps, across 5 API routes that orchestrate 4 external APIs behind one provider-agnostic call site.
- Measured the fact-checking engine at **93.3% (28 of 30 labelled claims) with zero incorrect verdicts** on an offline evaluation harness built with no new dependencies, by requiring every decisive verdict to carry resolved citations from two independent owner groups and routing every downgrade path to “unverified” instead of an adverse ruling.
- Cut a full run to roughly 13 model requests and held 18 of 18 concurrent analyses at HTTP 200, after reverse-engineering Google’s undocumented 15 requests/min ceiling from live 429 bodies and adding Gemini-to-Groq failover, a four-tier article-extraction fallback, and an SSRF guard on the user-supplied URL fetch.

Real-Time Conflict & Weapon Detection System

GitHub

Python, PyTorch, YOLOv8, Swin Transformer, OpenCV, CUDA

Mar 2025 – Sep 2025

- Built a three-stage cascaded vision pipeline (YOLOv8s person detection, Swin-Tiny conflict classifier, YOLOv8 weapon detector) over live 1280×720 video, cutting per-frame latency **2.2×** (228 to 104 ms on CPU, 59 to 26 ms on GPU at 38.4 FPS, RTX 3060) by gating the heavy stages behind the cheap one and scoring every third frame.
- Traced a reported 98.4% accuracy to a frame-wise split that left 98.5% of validation clips inside training, then re-established defensible baselines of **PR-AUC 0.808 and 74.8% per-video accuracy on 519 held-out videos** by regrouping with StratifiedGroupKFold on source video and pruning near-duplicate frames from 69,797 to 20,075.
- Set the threshold from the validation PR curve at **0.669 precision / 0.850 recall** and cut calibration error from 0.065 to 0.023 with temperature scaling; perceptual hashing exposed 48.6% cross-split contamination in the weapon dataset.

ctximg: Contextual Photo Search for PC

GitHub

Python, PyTorch, OpenCLIP, SigLIP 2, FastAPI, SQLite, NumPy

Oct 2025 – Mar 2026

- Built a fully offline photo search engine that embeds every image and every plain-English query into one CLIP vector space and ranks by cosine similarity with no tags, filenames, or filters, shipped as a tray-resident FastAPI desktop app with 19 REST endpoints, a dependency-free browser UI, and an 11-command CLI.
- Reached **100 / 49 / 32 images per second** across three selectable accuracy tiers (768-d at 224px up to SigLIP 2 1152-d at 384px) on a 6 GB RTX 3060, timed by the app on its own hardware, by moving inference to fp16 for a 3× speedup, scaling batch size to available VRAM, and recovering from CUDA OOM through recursive batch splitting.
- Made indexing incremental and crash-safe so an interrupt costs one batch rather than a full re-embed, using SQLite WAL with per-batch commits, path/size/mtime change detection, and threaded JPEG decode overlapped with GPU encode across 10 formats including HEIC; covered by **203 pytest tests green in 13 seconds**.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

Frontend: React, Next.js, Tailwind CSS, HTML, CSS

Backend & Databases: Node.js, Express, FastAPI, REST APIs, JWT Auth, MongoDB, MySQL, SQLite

AI / ML: PyTorch, OpenCV, CLIP Embeddings, Computer Vision, RAG, LLM APIs (Gemini, Groq)

Developer Tools: Git, GitHub, Docker, Vercel, Postman, Linux

ACHIEVEMENTS

Competitive Programming: LeetCode 300+ problems solved, contest rating 1600+

Web Developer, Aurora (Volunteer): Built and maintained an LPU student organisation’s website for a year.